UNIVERSITY OF LONDON

ST3189 Machine Learning Coursework

UOL Student Number: 210422078

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1 Supervised Learning (Regression)

1.1 Substantive Issue

Road transport is one of the biggest sources of pollution in the world. It's responsible for almost 30% of the carbon dioxide (CO2) released into the air globally (Lyft Urban Solutions, 2022).Public bike-sharing has become much more popular because it offers a greener and healthier way to get around, both for people and for the planet. We aim to create a model that can predict the number of bike rentals on a given day and identify the key factors that influence bike rental demand, such as weather, season, holidays, and day of the week.

1.2 Dataset Description & Variable

The dataset used in this project is the Bike Sharing Dataset. It includes daily records of bike rentals, along with information like date, weather, temperature, and whether it was a working day or holiday. The dataset has 16 columns in total, and the main value we want to predict is the total number of bikes rented in a day, called 'cnt'. There are no missing or repeated values in the dataset. A summary of the data is shown in Table 1 below:

	Name	Description
1	instant	Record index
2	dteday	Date
3	season	1=Spring 2= Summer 3= Fall 4=Winter
4	yr	Year (0=2011, 1=2012)
5	mnth	Month
6	holiday	0=not a public holiday ,1=public holiday
7	weekday	Day of the week
8	workingday	1 if neither weekend nor holiday, else 0
9	weathersit	1=Clear, Few clouds, 2=Mist + Cloudy, 3= Light Snow/Rain, 4=Heavy Rain/Snow
10	temp	Normalized temperature (actual temp = temp * 41 °C)
11	atemp	Normalized "feels-like" temperature (actual = atemp * 50 °C)
12	hum	Normalized humidity (0 to 1)
13	windspeed	Normalized wind speed (actual = windspeed * 67)
14	casual	Count of casual (non-registered) users
15	registered	Count of registered users
16	cnt	**Total count of rentals (`casual` + `registered`)** -target variable

Table 1: Data Variables

1.3 Evaluation Metrics

To measure how well each model performs, we use Root Mean Square Error (RMSE). RMSE tells us how close the model's predictions are to the actual number of bikes rented. A lower RMSE means the model is making better predictions, while a higher RMSE means the predictions are less accurate. The RMSE formula is:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(Predicted_i - Actual_i)^2}{n}}$$

1.4 Models

1.4.1 Linear Regression

Linear regression is a method that helps us see how two or more things are connected or affect each other.It helps predict the value of one variable (called the target) using other variables (called predictors). In our case, linear regression is used to predict the number of bike rentals (cnt) based on factors like temperature, humidity, windspeed, season, and whether the day was a holiday or a working day. The model works by fitting a line that best shows how the variables are connected. This line is calculated by finding the values that make the difference between predicted and actual values as small as possible.

In the data preparation stage standardized the dataset so that all variables were on the same scale. This helps the model treat all variables fairly and makes it easier to see which ones are most important. The data was divided so that 70% was used to train the model, and 30% was saved for testing. This will allow the model to be built using the training set and then evaluated on new, unseen data. To improve model performance, backward elimination was applied. This method begins with all available predictors and gradually removes the least useful ones. The process stops when further removal no longer improves the model, based on the Akaike Information Criterion (AIC). To better evaluate the model's ability to predict daily bike rentals, 10-fold cross-validation was carried out. This approach provides a more reliable estimate of model performance by testing it across different parts of the dataset.

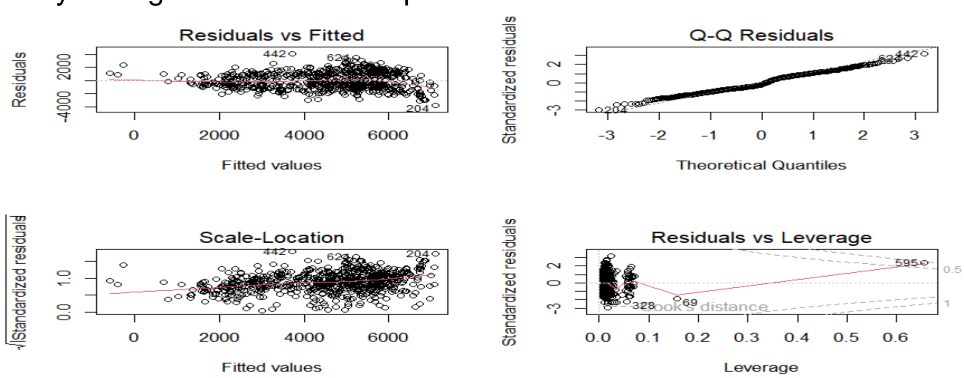

Figure 1: Diagnostic Plots

Plots were used to see if the linear regression model works well with the bike rental data. The Residuals vs Fitted plot (top left) shows some pattern, which suggests that the relationship between predictors and 'cnt' may not be perfectly linear. The Normal Q-Q plot (top right) shows that the residuals deviate from the straight line, indicating that the errors may not be normally distributed. Scale-Location plot (bottom left) has a slightly up red line, which a sign of heteroscedasticity. For Residuals vs Leverage plot, most points are fine, but one point (number 595) stands out. It has a lot of influence on the model. Some of the usual rules for linear regression may not be fully met. That does not mean the model is useless just that it might not be perfect. However, multicollinearity was not an issue, as all VIF values remained below 5. The RMSE for linear regression is 1311.4359.

1.4.2 Regularized Regression (Ridge Regression and Lasso Regression)

Regularized regression adds a penalty to the model to stop it from fitting the training data too closely. This helps avoid situations where the model works well on the training data but poorly on new data. Ridge and Lasso are two common methods that use different types of penalties. Lasso tends to shrink some coefficients exactly to zero, which means it can help select the most important features. Ridge shrinks coefficients towards zero but doesn't make them exactly zero. It keeps all features in the model but reduces their influence.

To find the best λ value, cross-validation was performed for both models. The Mean Squared Error (MSE) was plotted against log(λ). The graphs below show how the error changes as λ increases.

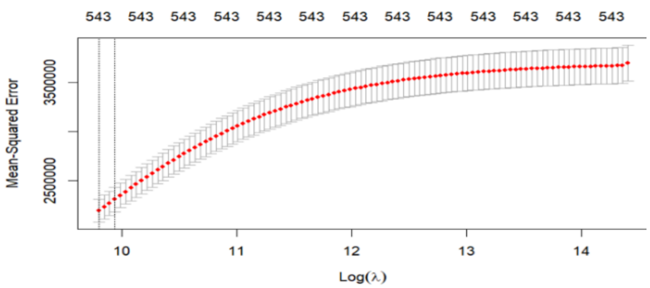

Figure 2: λ Selection via Cross-Validation for Ridge Regression

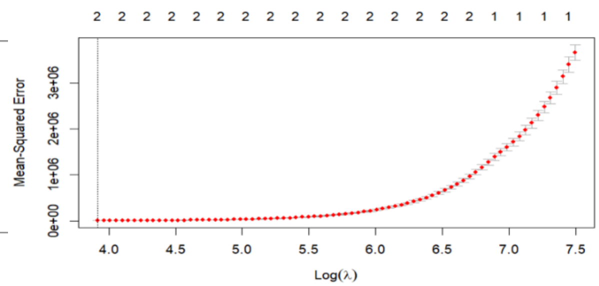

Figure 3: λ Selection via Cross-Validation for Lasso Regression

The Ridge model's error increases as λ increases. The best λ is found near the left, where the error is lowest. For Lasso, the error remains low over a wide range of λ values, but increases sharply at higher λ. The best λ is chosen at the point where the error is lowest. Ridge Regression achieved an average RMSE of 1481 across 10 folds. Lasso Regression achieved a much lower average RMSE of 61 across 10 folds. So, Lasso Regression performed significantly better on this dataset compared to Ridge Regression.

1.4.3 Classification & Regression Trees (CART)

CART is a type of decision tree that predicts results based on input data. It works by dividing the data into smaller groups using rules it learns from the data.The full regression tree was first trained with a complexity parameter (cp) set to 0, allowing the tree to grow to its maximum size.

To avoid overfitting, cost complexity pruning was applied using the cp value that gave the lowest cross-validation (CV) error. The pruned tree was then evaluated on the test set, producing a RMSE of 1338.52. To get a more stable performance estimate, 10-fold CV was applied to the CART model using the same pruning approach. Results of 10-fold CV were 1535, it is less accurate in predicting 'cnt' compared to other models. The final pruned tree is shown in Figure 4. The model highlights important decision rules. The root split shows the perceived temperature

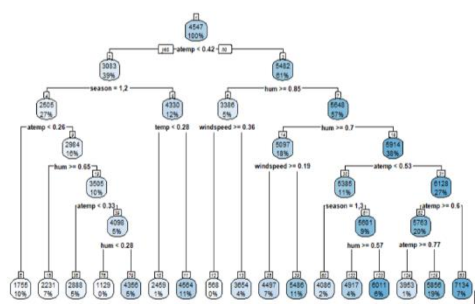

Figure 4: Pruned Cart Model

was the most important factor in the first decision. Other key splits include season, humidity, and wind speed, indicating how weather conditions affect bike rental behavior.

1.4.4 Random Forest (RF)

Random Forest is a machine learning method that creates many decision trees and combines their predictions to make the results more accurate and avoid overfitting. Each tree is trained on a random part of the data, and for each decision, a random set of factors is considered. This randomness helps the model work better on new data and reduces errors.A train-test split was used without scaling the data, as Random Forest is not sensitive to unstandardized inputs. The model achieved a RMSE of 1215.90 on the test set, which was lower than the RMSEs of Linear Regression and CART, but slightly higher than that of the Lasso Regression model. This shows that Random Forest is a strong and stable model for predicting bike rental demand. Moreover, Random Forest provides feature importance rankings. As shown in Figure 5, the three most important predictors were humidity (hum), temperature (temp), feels-like temperature (atemp).

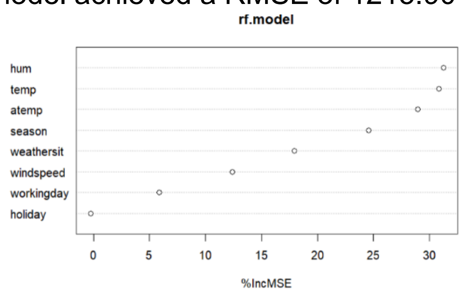

Figure 5: Variable Importance in Random Forest Regression

These variables had the largest increase in prediction error when removed from the model, indicating their strong influence on rental demand. This figure shows how much each variable contributes to prediction accuracy, based on the percentage increase in mean squared error (%IncMSE) when the variable is excluded.

1.5 Result

As shown in Table 2, Lasso Regression had the best performance with the lowest RMSE of 61. Despite a longer computation time, it outperformed other models. Random Forest followed with an RMSE of 1215.90, showing strong but slightly less accurate results. Linear Regression (RMSE 1311.44) and CART (RMSE 1535) performed moderately, while Ridge Regression (RMSE 1481) had the highest error. Lasso Regression is recommended for its accuracy, though its computational cost should be considered for larger datasets.

LR	Ridge	Lasso	CART	RF
1311.44	1481	61	1535	1215.90

Table 2: RMSE (test set error) values-models created

2 Supervised Learning (Classification)

2.1 Problem Statement

The Titanic disaster remains one of the most well-known tragedies in history, with over 1,500 lives lost (R.M.S Titanic - history and significance | National Oceanic and Atmospheric Administration, 2024). The goal of this project is to use the dataset to predict whether a passenger survived or not, based on variables such as age, gender, class, fare, and the port of embarkation.

2.2 Dataset Description

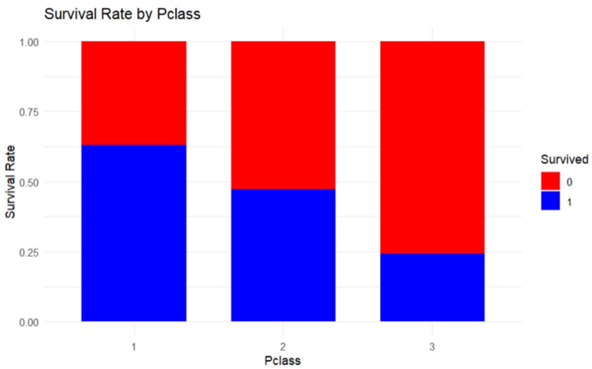

Figure 6: Survival Rate by Pclass

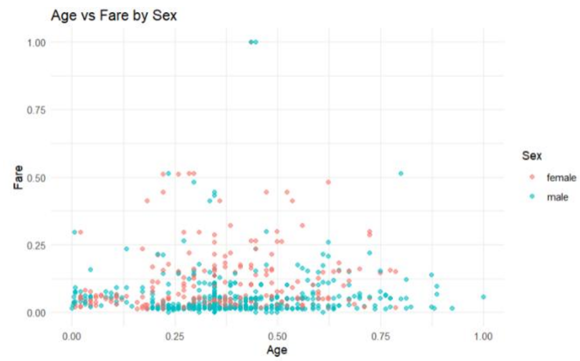

Figure 7: Age vs Fare by Sex

As shown in Figure 6, there is a clear relationship between passenger class and survival. Passengers in 1st class had the highest survival rate, shown by the larger blue section, while 3rd-class passengers had the lowest, showing how social class significantly affected survival chances during the Titanic disaster.

Figure 7 shows that women tended to have higher fares, which may suggest that they were more likely to be in the upper-class cabins. This, combined with the "women-first" principle during the rescue, demonstrates how both gender and class played a role in determining survival chances.

2.3 Evaluation Metrics

$$Precision = \frac{True\ Positive}{True\ Positive + False\ Positive} \quad Recall = \frac{True\ Positive}{True\ Positive + False\ Negative} \quad F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Since the dataset is nearly balanced, we will use two main metrics to evaluate the model which are F1 Score and ROCAUC. The F1 Score combines two important factors, precision and recall, to give a more complete measure of the model's performance. This is especially useful for datasets like ours, where the classes are almost equal. The ROCAUC is our second metric, and it helps us understand how well the model can distinguish between positive and negative cases at different levels of confidence. In the confusion matrix, True Negatives are cases where the model correctly predicts a passenger did not survive, True Positives are cases where it correctly predicts a passenger survived, False Positives are when it wrongly predicts a passenger survived when they didn't, and False Negatives are when it misses a passenger who survived.

2.4 Models

2.4.1 Logistic Regression

Logistic regression is used to predict the outcome of a binary variable, such as whether a passenger survived or not on the Titanic, based on other input variables. It calculates the probability of the outcome happening. For data preparation, the target variable Survived was transformed from a binary variable (0 or 1) into a factor with two levels. A train-test split was used to separate the data for training and testing the model.

While building the logistic regression model, we encountered errors like "algorithm did not converge" and "fitted probabilities numerically 0 or 1." This suggests that the data may be perfectly separable. In simpler terms, there might be a clear distinction between the two classes (survived or not) that the model couldn’t handle well. When this happens, logistic regression struggles to make accurate predictions. However, other classification models, such as KNN, SVM, and Neural Networks, can still work well in such situations to make predictions.

2.4.2 K-Nearest Neighbours (KNN)

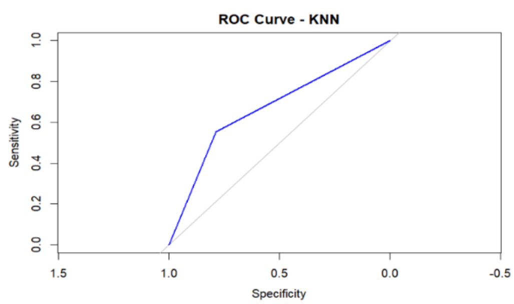

Figure 8: KNN ROC

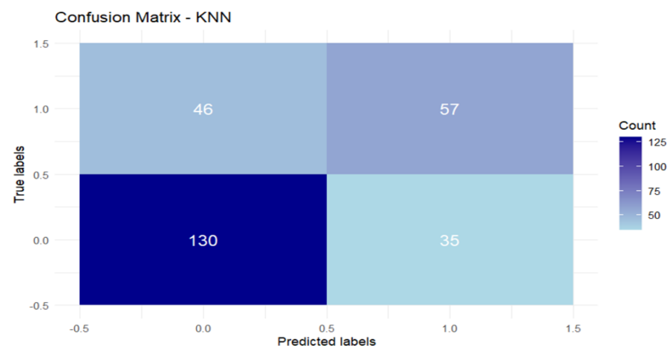

Figure 9: KNN Confusion Matrix

KNN uses the distance between data points to classify new instances. The model showed decent performance, with an F1 score of 0.585, an accuracy of 0.698, and an ROCAUC score of 0. 671.The ROC curve indicates moderate ability to distinguish between the two classes. In figure 9, the confusion matrix shows how well the KNN model performed. It correctly predicted 46 True Positives, 130 True Negatives, but had 57 False Positives and 35 False Negatives. KNN model may not be a suitable predictive model.

2.4.3 Support Vector Machine (SVM)

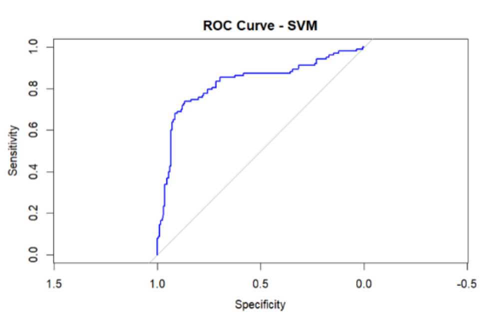

Figure 10: SVM ROC

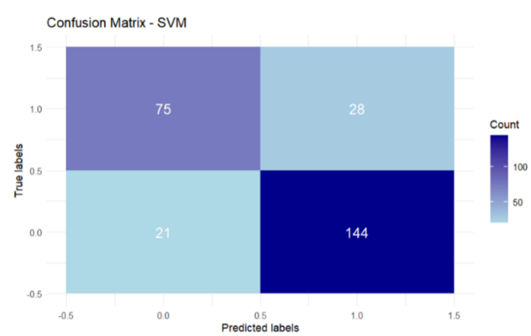

Figure 11: SVM Confusion Matrix

SVM is a machine learning algorithm that looks for the best boundary (called a hyperplane) to separate data into two different groups.The model achieved an F1 score of 0.204 and a ROCAUC score of 0. 827. In figure 11, the confusion matrix shows the results of the SVM model. It correctly predicted 75 True Positives (survived) and 144 True Negatives (did not survive), with 28 False Positives and 21 False Negatives. Compared to the KNN model, the SVM model has better performance with fewer False Negatives, but there is still room for improvement in predicting survivors.

2.4.4 Neural Network (NN)

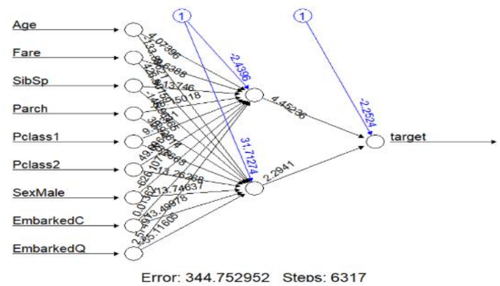

Figure 13: NN diagram

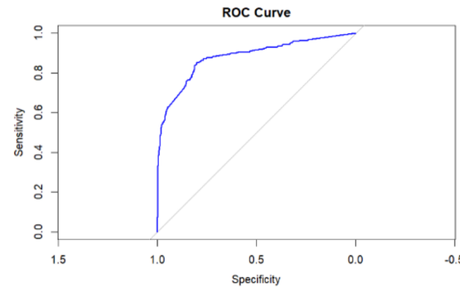

Figure 12: Neutral Network ROC

Networks (NN) are designed to recognize patterns by simulating the way the human brain works. The model learns from input data and adjusts internal parameters (weights) to make predictions. For this classification task, the model achieved an F1 score of 0.77 and a ROCAUC score of 0.881, showing strong performance. This diagram illustrates the architecture of the neural network used for the classification task. It shows the input features, hidden layer, and the output layer, which predicts whether a passenger survived or not.

2.5 Recommendation

The Neural Network (NN) model appears to be the best, based on its higher F1 score (0.77) and ROCAUC score (0.881). It shows better performance than the other models in both classification accuracy and ability to separate classes.

Classifiers	F1	ROCAUC	Accuracy
K-Nearest Neighbors	0.585	0.671	0.698
Support Vector Machine	0.204	0.827	0.183
Neutral Network	0.77	0.881	0.814

Table 3: Summary of classification

3 Unsupervised Learning

3.1 Problem Statement

This dataset includes customer details such as gender, age, annual income, and spending score. By analyzing these attributes, we aim to identify patterns or groups within the customers. This will help businesses to understand different customer segments, allow them to tailor marketing strategies for optimizing services.

3.2 Dataset Description

Figure 14: Correlation Matrix

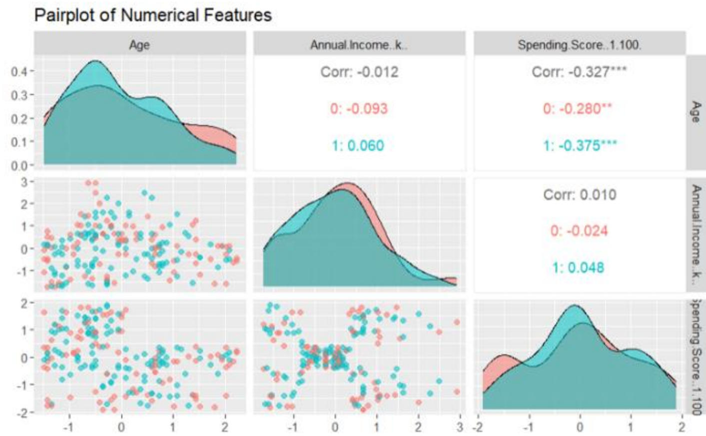

Figure 15: Pair plot of unsupervised learning

In Figure 14, Age and Spending Score have a moderate negative correlation of -0.33. This indicates that as age increases, spending tends to decrease. Annual Income has a very weak correlation with both Age and Spending Score, suggesting that income does not strongly influence these factors. This insight is important in understanding consumer behavior in the dataset.

Figure 15 provides the distribution plots of the features. The distribution of Age and Annual Income shows a mostly right-skewed distribution, meaning most customers are younger and have a lower income. However, the Spending Score appears to have a more even distribution, reflecting diverse spending habits among the customers. These findings highlight how different factors such as income and age influence consumer spending patterns, with a noticeable trend toward younger individuals and lower-income groups.

3.3 Models

3.3.1 Principal Component Analysis (PCA)

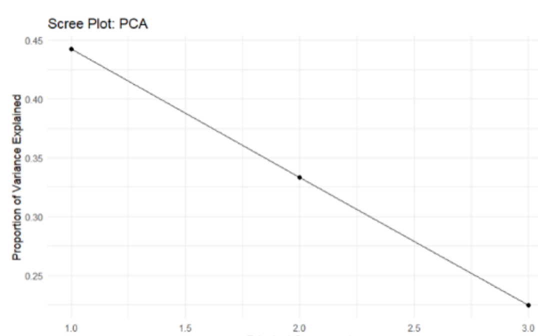

Figure 16: PCA Scree Plot

PCA is a technique that reduces the number of variables in a data set while keeping the important information. It changes a set of related variables into a smaller set of unrelated ones called principal components. These principal components help identify patterns and relationships in the data. In figure 16, the first three components explained 44.3%, 33.3%, and 22.4% of the variance, respectively, together accounting for 100% of the total variance. By reducing the dimensionality to just the first two components, which capture 77.6% of the total variance, this can simplify the model and prevent overfitting, without losing significant information from the data.

3.3.2 K-Means Clustering

K-Means clustering is a technique used to group data into clusters based on similarities between the data points. We used the elbow method to choose the number of clusters, as shown in Figure 17, where the "elbow" point suggests three clusters are ideal for our analysis. This is where the sum of squared errors decreases and stabilizes, making it the best number of clusters to analyze.

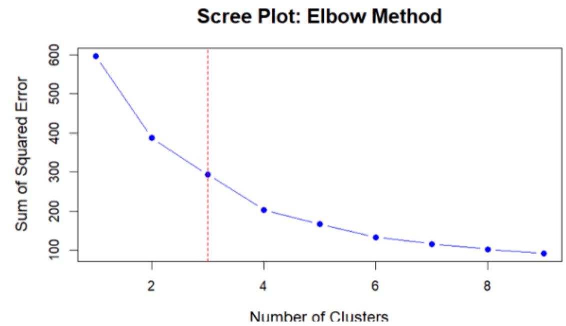

Figure 17: K-Means Clustering

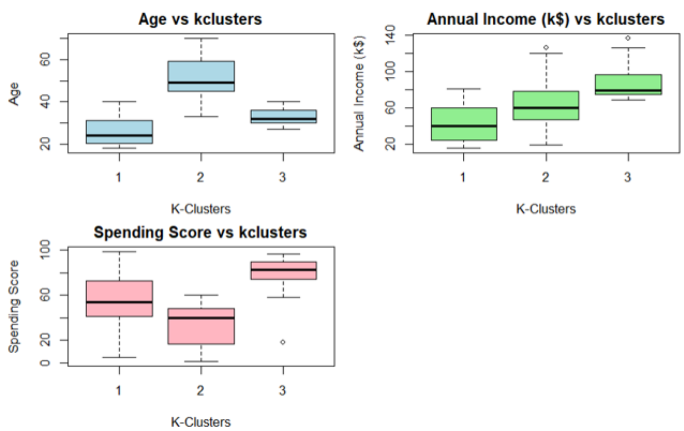

Figure18 : Variables vs Clusters

The boxplots in Figure 18 show how each cluster is distinguished by important features like Age, Annual Income, and Spending Score. For example, Cluster 1 has the youngest age group with the lowest income, while Cluster 3 has the highest income and spending score. These patterns help us understand how customers can be segmented based on these factors, revealing useful insights for targeting specific groups. Based on these findings, it is reasonable to categorize Cluster 1 as '**Young Middle-class Spenders**, Cluster 2 as '**Conservative Low-spending Group**, Cluster 3 as '**High-income High-spending Elite**.

3.3.3 Hierarchical Agglomerative Clustering (HAC)

Hierarchical clustering is a method that groups similar data points into clusters in a tree-like structure. It works by combining the closest clusters together until all points are in one cluster or a stopping condition is met. This process can be shown in a diagram called a dendrogram, which illustrates the similarities between clusters and how they are step by step. To measure how similar the clusters are, different methods like single, average, complete, and Ward's linkage can be used. For our data, we found that Ward's linkage worked the best, providing the strongest grouping of data points. This means that Ward's method was the most suitable for our analysis.

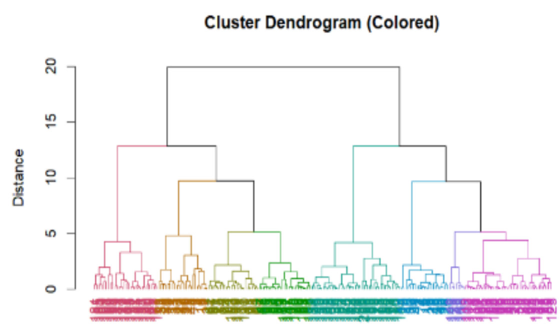

Figure 19: Cluster Dendrogram

From this analysis, we identified an optimal number of clusters by looking at both the Silhouette and Gap statistics, which suggested 8 clusters for the most robust result as stopping criterion.

```
[1] "Cluster distribution for k = 7:"
hc.cut.tree.sil
 1  2  3  4  5  6  7
22 21 38 45 39  7 28
[1] "Cluster distribution for k = 8:"
hc.cut.tree.gap
 1  2  3  4  5  6  7  8
22 21 38 22 23 39  7 28
```

Figure 23: cluster plot

Figure 23 shows 8 clusters identified by the HAC method, each cluster represented by different colors and shapes. The first two principal components, Dim1 (44.3%) and Dim2 (33.3%), explain 77.6% of the variation in the data. The plot clearly shows that the groups are well-separated, meaning the HAC algorithm has successfully grouped customers with similar characteristics. Cluster 1 (red) might represent high-spending, high-income customers. Cluster 2 (yellow) may represent low-income, low-spending customers. Clusters in the middle could represent customers with moderate income and spending, or more complex behavior that varies by other features.

3.4 Result

kclusters <fctr>	CustomerID_mean <dbl>	Gender_mean <dbl>	Age_mean <dbl>	Annual.Income..k.._mean <dbl>	Spending.Score..1.100._mean <dbl>
1	60.6	0.588	25.8	42.8	53.6
2	102.6	0.549	51.3	61.8	34.2
3	162.0	0.537	32.9	87.3	80.0

3 rows

Figure 21: K-Means Cluster Recommendations

A tibble: 8 × 4

hclusters <fctr>	avg_age <dbl>	avg_income <dbl>	avg_spending <dbl>
1	44.3	25.8	20.3
2	24.8	25.6	80.2
3	28.2	54.3	50.7
4	63.9	53.0	50.6
5	49.2	57.4	46.2
6	32.7	86.5	82.1
7	22.7	74.7	19.3
8	43.9	91.3	16.7

8 rows

Figure 22: HAC Recommendations

In this analysis, I applied K-Means clustering and HAC to segment customers based on their age, annual income, and spending score. K-Means clustering divided the customers into three distinct clusters, each representing a different spending behavior. Cluster 1 represents the Young Middle-class Spenders, a group with moderate income and spending, likely responsive to affordable products and promotions. Cluster 2, labeled as the Conservative Low-spending Group, consists of individuals with higher income but lower spending. This group could benefit from exclusive products or luxury offers to encourage higher engagement. Cluster 3 represents the High-income High-spending Elite, ideal for targeted premium products and personalized services due to their high engagement and purchasing power.

The HAC analysis created eight clusters, offering a finer segmentation of customers. The higher number of clusters allowed for a more detailed understanding of customer behaviors, highlighting sub-groups that might not be captured in the broader K-Means categories. For instance, certain HAC clusters like the young, high-income, high-spending group require specific targeting with trendy, high-end products, while others with high income but low spending need targeted incentives to increase their purchasing behavior. By understanding the characteristics of each cluster, businesses can tailor their marketing strategies, whether through promotions for middle-class spenders or luxury offerings for high-income groups to maximize customer engagement and sales.

References

Bommae Kim. “Understanding Diagnostic Plots for Linear Regression Analysis.” *Understanding Diagnostic Plots for Linear Regression Analysis | UVA Library*, 21 Sept. 2015, library.virginia.edu/data/articles/diagnostic-plots.

Lee, Kevin, et al. “What Exactly Does Model_matrix() Do?” *Stack Overflow*, 1 Apr. 1965, stackoverflow.com/questions/62560713/what-exactly-does-model-matrix-do.

Lumumba W. Victor. “Sign In.” *RPubs*, 10 Apr. 2023, rpubs.com/lumumba99/1026665.

R.M.S Titanic - History and Significance | National Oceanic and Atmospheric Administration, 18 July 2024, www.noaa.gov/office-of-general-counsel/gc-international-section/rms-titanic-history-and-significance.

Team, DataCamp. “Hierarchical Clustering in R: Dendrograms with Hclust.” *DataCamp*, DataCamp, 24 July 2018, www.datacamp.com/tutorial/hierarchical-clustering-R.

“What Is a Bike Share Program and How Does It Work?” *Lyft Urban Solutions*, 17 Jan. 2022, lyfturbansolutions.com/blog/2022/01/what-is-a-bike-share-program-and-how-does-it-work.

Dataset link:

1.regession

https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset

2.classification

https://www.kaggle.com/c/titanic/data

3.unsupervised learning

https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python?resource=download